

Faunal Diversity and Ecology of Moss Inhabiting Terrestrial Invertebrate Fauna of Schirmacher Oasis, Antarctica

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ABSTRACT

Zoological Survey of India (ZSI) participated in XXIV Indian Antarctic Expedition during (2004-2005). The present research work was carried out during the austral summer of January-February 2005, in different localities of Schirmacher Oasis to understand the faunal diversity of soils and moss inhabiting invertebrate fauna along with some physico-chemical parameters like soil temperature and relative humidity. In this study, 5 species of Protozoa, 5 species of Nematoda, 2 species of Collembola, 5 species of Mites and 1 species of Rotifera including some new records from Schirmacher Oasis are being reported.

INTRODUCTION

Antarctica is the last terra incognita and terrarium on the earth, covered with snow and ice that surrounds the south pole. The name of region originated from the Greek word 'Antarctikas' which means 'opposite' to bear the northern constellation i.e. the opposite to arctic. The Indian Antarctic station, Maitri is located in the Schirmacher Oasis in the Dronning Moud Land, East Antarctica (117 m above mean sea level). The nearest steep cliff of the east west trending glacier on the southern side of the station is more than 700 m away from the station and 300 m high. The snow covered surface during summer season is more than 0.5 km away from the station, the surface around the Maitri station is covered by sandy and loamy sand type of soil. Schirmacher Oasis is one of the small ice free polar rocky hill with a maximum width of 2 km and length of about 18 km in east-west direction. Its area is approximately 35 sq km. There are over more than 100 fresh water lakes of different shapes and sizes in the Oasis.

Early Study on Invertebrate Fauna of Schirmacher Oasis

Protozoa : Ingole and Parulekar (1987), were the first to report Protozoa, from this region and recorded only one species of ciliate, viz., *Oxytricha fallax* Stein and noted that Protozoa are the most dominant fauna of this oasis, comprising 22.31 percent of the total lacustrine microfauna. Afterwards Mitra *et al.*, (1997) and Mitra (1999) reported 17 species from this area of which 16 were new addition. Chatterjee *et al.*, (2000) made the community analysis of testacean species from this cold desert for the first time.

Nematode : Hazra (1990) was the first to record 5 genera/species of nematodes from the Schirmacher Oasis. Addition of another 5 genera/species of nematodes from this oasis was made by Mitra (1997). A substantial contribution to knowledge of the diversity, ecology and population fluctuation of nematodes was made by Hazra (1994) and Hazra and Mitra (2002). Ghosh *et al.*, (2000) observed an unusual type of nematode distribution in different lake water systems of Schirmacher Oasis. It was also observed that maximum number of nematodes was distributed in between shelf and polar ice cap whereas minimum in the lakes nearer to shelf. The first record of nematode, *Antarctenchus motililus*, a new species, was made by Ghosh *et al.*, (2005) from the Schirmacher Oasis. The samples were brought during XV and XXII Indian Scientific Expedition to Antarctica.

Rotifera : Venkataraman (1998) first recorded one species of rotifer *Philodina gregaria* from Schirmacher Oasis and later Mitra (1999) also reported the same species from the Oasis.

Collembola : Mitra (1999) has first reported Collembola from this Oasis. Two species of two families viz., Isotomidae and Entomobryidae have been reported from this zone.

Acarine : Mitra (1999) reported two unidentified mite species from Schirmacher Oasis. Afterwards Sanyal *et al.*, (2002) have described 3 new species of oribatid mites from this area.

MATERIALS AND METHODS

Geo-position

Latitude : 70°44'30" - 70°46'30" S

Longitude : 11°22'44" - 11°22'44" E

A total of 24 sites (21 lakes, 2 swampy areas, and 1 wastewater site) were surveyed during 2004-2005 austral summer. Maximum number of samples was collected from the Priyadarshini Lake at the periphery of near by melt water stream of Maitri. Soil samples were collected from the stagnant water pool behind the Maitri station, which receive kitchen waste water from Maitri. A total of 100 soil samples were collected at random from 25 different sites at the rate of 4 samples from each site, by using metal shovel and stored in polythene bags. The bags containing samples were tightly closed with rubber bands and kept in natural atmosphere in Antarctica. During the return voyage the samples were stored in cold storage in the ship to avoid evaporation. Some freshly collected moss samples were examined under binocular after extraction of moss samples with the help of Tullgren funnel soil extraction apparatus in the laboratory inside the summer hut of the Schirmacher Oasis. Remaining soil samples were processed for the isolation of mites, Collembola, Protozoa, Nematodes, Rotifera and Tardigrada in the laboratory in our institute.

RESULTS

Protozoa : Protozoa are single-celled eukaryotic organisms which are, with some exceptions, visible only with the aid of a microscope. These organisms mostly range from 5 µm to 250 µm. A huge number of Protozoa are known to occur in most habitats on moss, peat, mineral soil and animal guano in Antarctica. Smith (1978) has recorded 124 species of Protozoa from the Antarctic and Marine Antarctic islands. Ingole and Parulekar (1987) initially recorded Protozoa represented only by one species of Ciliate, i.e. *Oxytricha fallax* Stein.

In the present study 6 genera/species were recorded viz., *Arcella arenaria* Greff, *Arcella catinus* Penard, *Assulina muscorum* Greff, *Corythion dubium* Taranek, *Oxytricha fallax* Stein and *Callopoda* sp. (**Table 1**) (Mitra 1999).

Rotifera : The rotifers are commonly known as “rotatoria” or “wheel animalcule”. *Philodina gregaria* were obtained from 9 different sites (**Table 1**) and was widely distributed in Schirmacher Oasis.

Collembola : During the present study Collembola, represented by two species viz., *Isotoma* sp. and *Cryptopygus* sp. were collected from 6 sites.

Table 1

Sites		Protozoa	Rotifera	Collembolan	Acarina	Nematode	Samples
Lake	1						00
	2	+	+				02
	3	+	+	+		+	04
	4	+			+	+	03
	5			+	+		02
	6	+	+		+	+	04
	7						00
	8						00
	9						00
	10	+	+	+			03
	11	+					01
	12						00
	13	+	+				02
	14						00
	15						00
	16	+	+				02
	17	+	+	+	+	+	05
	18						00
	19						00
	20		+	+		+	03
	21	+				+	02
Polluted site	22			+	+		02
Swampy areas	23	+	+			+	03
	24	+					01
Open land	25						00
(Total)		12	09	06	05	07	39

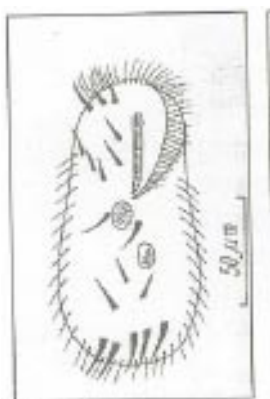
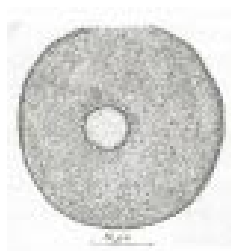
Nematodes

Nematodes are one of the most important groups of invertebrate animals occurring in the all types of habitats such as soil of lakes, rivers and deep oceans. Some nematodes can also be found in the rocky soil or in hot springs soil under high temperature. On the other hand, some nematodes can thrive even in the coldest soil where temperature varies from -89° to +5°C. The nematode fauna encountered in this study belonging to 5° genera and obtained from different localities. It was found that the *Antarctenchus* sp. was the most dominant species of the area surveyed at Schirmacher Oasis. The following nematode species were collected from the Moss turf and soil samples from the periphery of various lakes and studied up to generic level viz., *Antarctenchus* sp., *Helicolenchus* sp., *Dorylaimida* sp., *Monochus* sp., *Rhabdidis* sp., (Ghosh *et al.*, 2005).

Acarina : During the present study 5 species under 5 genera of mites were recorded from 5 different sites of Schirmacher Oasis. They were mainly collected from the polluted sites and peripheral areas of Priyadarshini lake located at Schirmacher Oasis viz. *Suidasia nesbittii*, *Tyrophagus longior*, *Haplochthonius simplex*, *Haplochthonis antarcticus* and *Hypoaspis* sp. (Figs. 1-19) (Sanyal 2002).

DISCUSSION

Review of the earlier works done on invertebrates of Schirmacher Oasis reveals that 5 groups of invertebrates viz., Protozoa, Nematoda, Collembolla, Mite and Rotifera have been recorded from here. During the present study, one hundred samples were collected from different areas in the Schirmacher Oasis. Five microscopic invertebrate fauna (microbial fauna – microbial flora is heard not microbial fauna. These are not microbial but microscopic fauna) were collected from this area such as Protozoa, Nematode, Rotifera, Collembolla and Mites from 15 different sites. The invertebrate fauna obtained from all the sites of Schirmacher Oasis are given in **Table 1**. Protozoans were dominant, and represented by 7 genera/species and those like *Parmulina* sp., *Corythion dubium*, *Oxytricha fallax*, *Assulina muscorum*, *Arcella arenaria* have been widely distributed in different lakes of Schirmacher Oasis. *Parmulina* sp. have also been reported from different parts of the world including the Himalayas. Maximum number of 5 invertebrate groups were obtained from the Lake no. 17, followed by 4 groups from Lake no. 3 and 6. The Acarine and Collembolla were extracted collected in low number.

Fig. 1 : *Oxytricha fallax*Fig. 2 : *Stylonchia* sp.Fig. 3 : *Colpoda* sp.Fig. 4 : *Arcella arenaria*Fig. 5 : *Arcella* sp.Fig. 6 : *Difflugia* sp.Fig. 7 : *Assulina muscorum*Fig. 8 : *Carythion dubium*

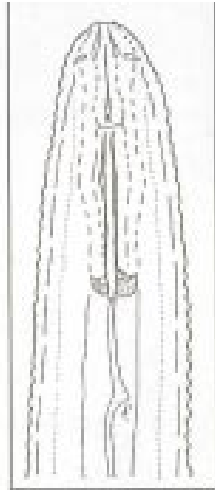


Fig. 9 : *Antarchtenchus motilitus* (anterior end)

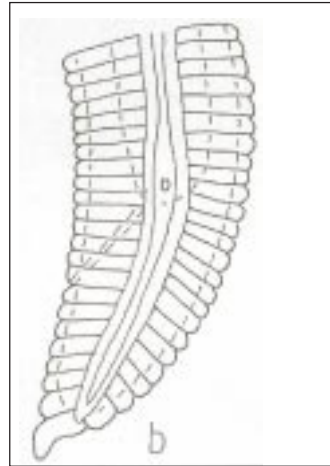


Fig. 10 : *Antarchtenchus motilitus* (tail)



Fig. 11 : *Haplochthonius antarcticus*



Fig. 12 : *Haplochthonius maitri*

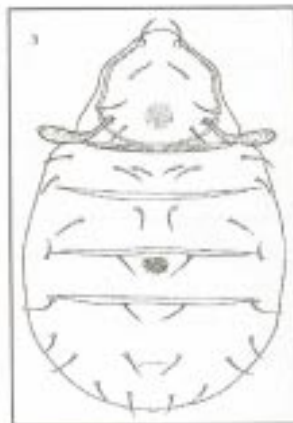


Fig. 13 : *Haplochthonius* sp.



Fig. 14 : Moss samples collected from various sites in Schirmacher Oasis



Fig. 15 : South Polar Skua near Maitri



Fig. 16 : Sampling from a lake in Schirmacher Oasis



Fig. 17 : ZSI Camp laboratory, Tirumala Summer Hut, Maitri



*Fig. 18 : Tullgren Funnels in operation in camp laboratory,
Tirumala Summer Hut, Maitri*



Fig. 19 : Sorting out of invertebrate groups under binocular

Rotifera was collected from 9 sites and Acarina was obtained from 5 sites and mainly from polluted sites and at the periphery of Priyadarshini Lake. Collembolla was found in 6 sites (**Table 1**). The Nematode fauna encountered in this study belonged to 5 genera and obtained from 5 different sites, of which *Antarctenchus* sp. was the most dominant species. The maximum Nematode were collected at the end of January when the surrounding temperatures were high and moss vegetation was in full grown stage. Thereafter, population started declining and reached to a minimum number in the last week of February, when low temperature was recorded. In general it is evident that maximum number of invertebrate groups was obtained from around the Priyadarshini Lake. Maslen (1979, 1981) also observed maximum population of nematodes during austral summer and low in winter. Ingole and Parulekar (1987) have also reported the similar results from Schirmacher Oasis.

It is evident from the Table 1 that 9 sites (1, 7, 8, 9, 12, 14, 15, 18, 19, 25) were devoid of any faunal components during the present study. This may be due to the presence of poor vegetation cover as most of these sites were nearer to polar ice caps. Block (1966) suggested that direct correlations exist between the vegetation cover of microflora and Acarina. Maslen (1981) observed that maximum aggregation of population in the upper layer during summer in Antarctic region might be related to the density and texture of moss which acts as a source of food for nematodes.

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REFERENCES

1. Chatterjee, P., Sinha, J., Das, A.K., and Mitra, Bulganin. 2000. Studies on some Antarctic Testacians (Protozoa : Rhizopoda) : A community analysis. *Ecol. Env. & Conc.*, **6** (2): 237-243.
2. Ghosh, S.C., Mitra, Bulganin and Chatterjee, A. 2000. Distribution of Nematode fauna in different lake water system in Schirmacher Oasis in East Antarctica. *J. Interacad*, **4** (3): 420-425.
3. Ghosh, S.C., Chatterjee, A. Mitra, Bulganin., and Dey, J.K. 2005. *Antarchtenchus motililus* sp. n. Nematode : Tylenchida) from Schirmacher Oasis, East Antarctica. *Intarcademica*, **9** (3) : 367-371.
4. Hazra, A.K. 1990. Terrestrial invertebrate fauna in Antarctica. In: *Taxonomy in Environment Biology* (ed. Director, 281) pp. 229-242.
5. Hazra, A.K. 1994. A study of population ecology of soil nematode fauna in relation to some Schirmacher Oasis, Antarctic Scientific report. DOD, Tech. Pub., No.6: 65-90.
6. Hazra, A.K. 1998. The diversity and colonization of terrestrial fauna of Schirmacher Oasis, East Antarctica. Abstract: SCAR VII Biology symposium, New Zealand.
7. Hazra, A.K. and Mitra, Bulganin. 2002. Diversity & Colonization of the terrestrial invertebrate fauna at Schirmacher Oasis, East Antarctica. *Rec. Zool. Surv. India*, **100** (3-4): 145-159.
8. Ingole, B.S. and Parulekar, A.H. 1987. Microfauna of Schirmacher Oasis, Antarctica : I. Water moss communities, Scientific report of Indian expedition to Antarctica. Tech. Pub. No. 4, pp. 139-148.
9. Maslen, N.R. 1979. Addition to the nematode fauna of the Antarctic region with key taxa. *Bull. Brit. Antarct. Surv.* **49**: 207-229.
10. Maslen, N.R. 1981. The signey island terrestrial reference soil: XII. Population ecology of nematodes with additions to the fauna. *Bull. Antract. Bull., Br. Antarct. Bull.*, **53**: 57-75.
11. Mitra, Bulganin., Das, A.K. and Chatterjee, P. 1997. Protozoan habitats in the moss habitats of fresh water lake at Schirmacher Oasis. In : Proc. International Symp. On sustainable utilization of biodiversity. Dehradun (Abstract published :18-19).
12. Mitra, Bulganin. 1999. Studies on Moss inhabiting invertebrate fauna of Schirmacher Oasis. Scientific report, DOD., Tech. Pub. No. 13: 93-108.
13. Sanyal, A.K. Basak, S and Barman, R.P. 2002. Three new species of Oribatid mites (Acarina, Oribatida : Haplothoniidae) from the Antarctic continent. *Acarina*, 10(1): 57-63(Russia).
14. Sanyal, A.K. 2005. Diversity of invertebrate fauna of Schirmacher Oasis, East Antarctica. DOD. Tec. Pub. No. 17, pp.173-187.

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15. Sanyal, A.K. 200. A study of ecology of soil mite in two contrasting sites at Schimacher Oasis, East Antarctica. *Acarina* (Russia).
16. Venkataraman, K. 1998. Studies on Phylum Tardigrada and other associated fauna South polar Skua and Bird and Mammal logging 1994-1995 expedition. Department of Ocean Development, Tech. Pub. No. 12: 221-243.